

March 19, 2026

Dr. Just Cebrian, Dr. Paul Montagna, Dr. Melissa Wong, Editors-in-Chief
Estuaries and Coasts

Dear Dr. Cebrian, Dr. Montagna, and Dr. Wong,

Enclosed please find our manuscript, titled 'Managing the unmanageable: Understanding the competing effects of salinity and nutrient loading on estuarine water quality', to be considered as a management applications article in Estuaries and Coasts.

Previous work we published in Estuaries and Coasts demonstrated a long-term reduction in salinity in Tampa Bay, Florida (<https://doi.org.10.1007/s12237-024-01385-0>). The objective of that work was to assess the trends and establish associations with recent seagrass losses. These changes also have significant implications for the management of water quality. In this work, we use empirical models to identify unique effects of salinity and nutrient loading on chlorophyll-a concentrations to assess how large-scale freshening of the bay may limit the effectiveness of nutrient management efforts. In addition to quantifying how both have influenced chlorophyll-a concentrations over time, we provide simulation analyses to assess the likelihood of attaining regulatory standards in the future under changing baselines. Overall, we provide a cautionary tale that climate-related stressors can confound existing management models, particularly related to nutrient management, and that these effects should be considered in future management efforts.

This work has not been published elsewhere, nor is it under consideration for publication anywhere else. Each author has contributed substantially to this work. All authors have contributed to the writing and review of the original draft. Analysis was conducted by the first author, with interpretation of results supported by all other co-authors. All authors have approved the final draft and the submitted product was prepared according to instructions for authors provided on the journal's website.

We greatly appreciate the opportunity to publish this work in Estuaries and Coasts and are confident readers will consider it a valuable contribution. Please feel free to contact us directly should additional information be needed.

Regards,

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